

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("netflix_titles.csv")

df.head()
```

```
Out[2]:
```

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021

```
In [3]: print("Rows and Columns:", df.shape)
```

Rows and Columns: (8807, 12)

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 8807 entries, 0 to 8806  
Data columns (total 12 columns):  
#   Column          Non-Null Count  Dtype  
---  ---  
0   show_id         8807 non-null   object  
1   type            8807 non-null   object  
2   title           8807 non-null   object  
3   director        6173 non-null   object  
4   cast            7982 non-null   object  
5   country         7976 non-null   object  
6   date_added      8797 non-null   object  
7   release_year    8807 non-null   int64  
8   rating          8803 non-null   object  
9   duration        8804 non-null   object  
10  listed_in       8807 non-null   object  
11  description      8807 non-null   object  
dtypes: int64(1), object(11)  
memory usage: 825.8+ KB
```

In []:

In [5]: `df.describe()`

Out[5]:

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

In [6]: `df.isnull().sum()`

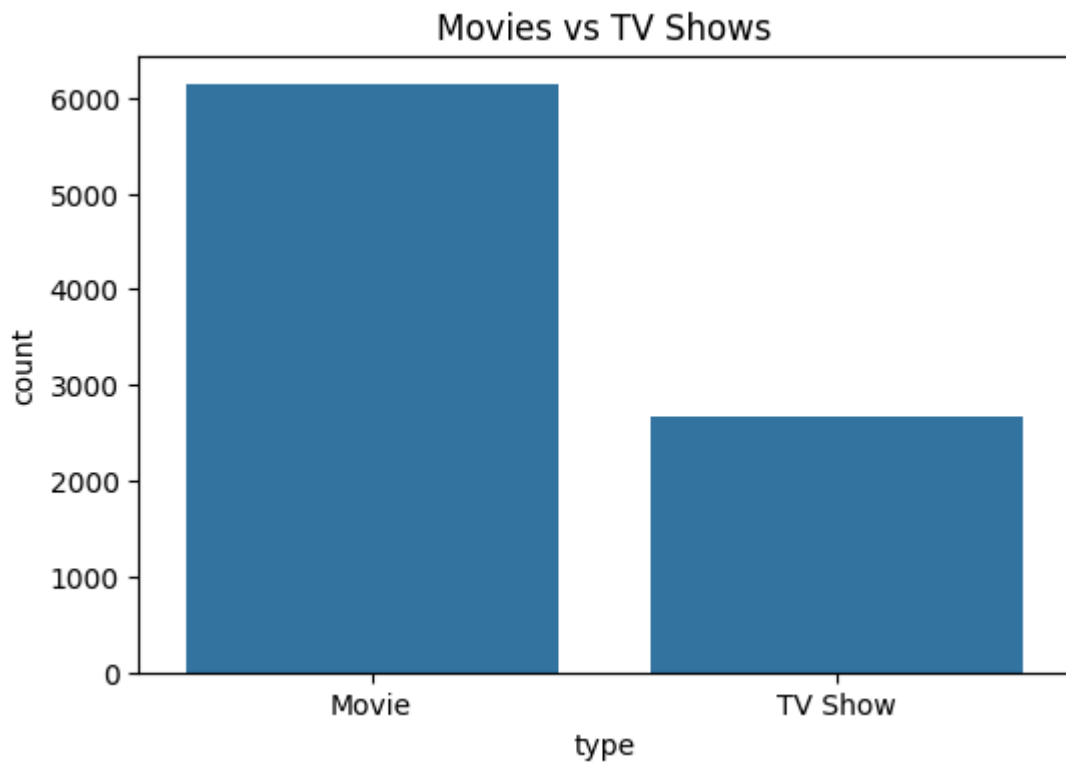
Out[6]:

show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

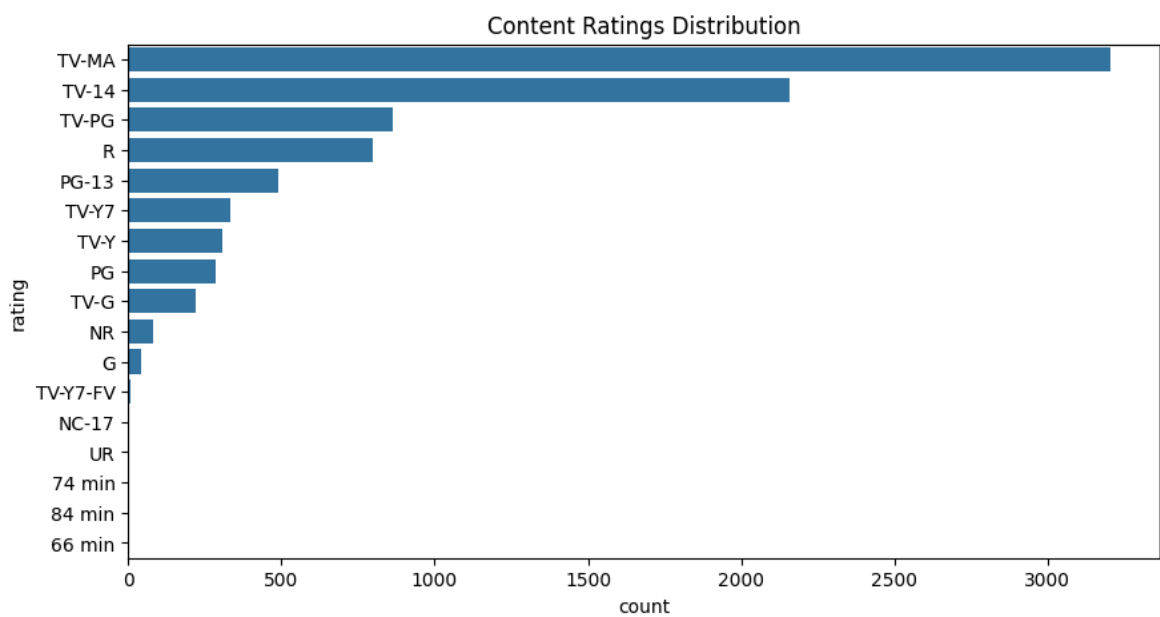
In [7]: `plt.figure(figsize=(6,4))
sns.countplot(x='type', data=df)`

```
plt.title("Movies vs TV Shows")
plt.show()
```



```
In [8]: plt.figure(figsize=(10,5))
sns.countplot(y='rating',
              data=df,
              order=df['rating'].value_counts().index)

plt.title("Content Ratings Distribution")
plt.show()
```

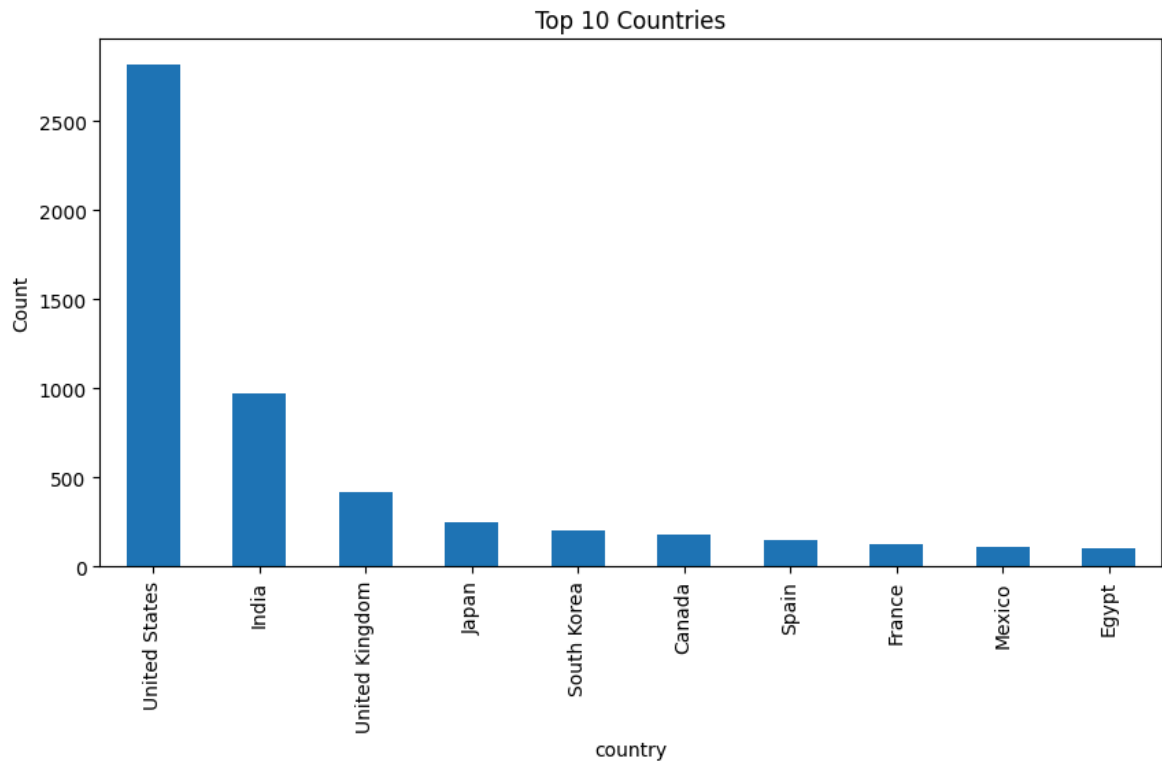


```
In [9]: plt.figure(figsize=(10,5))

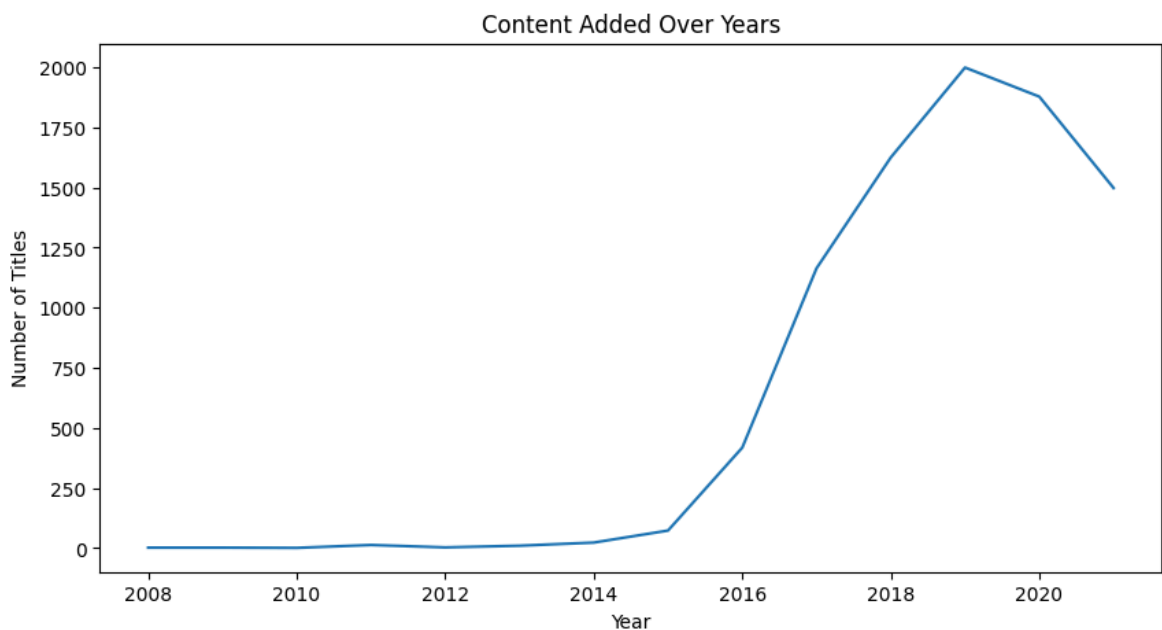
df['country'].value_counts().head(10).plot(kind='bar')

plt.title("Top 10 Countries")
```

```
plt.ylabel("Count")  
plt.show()
```



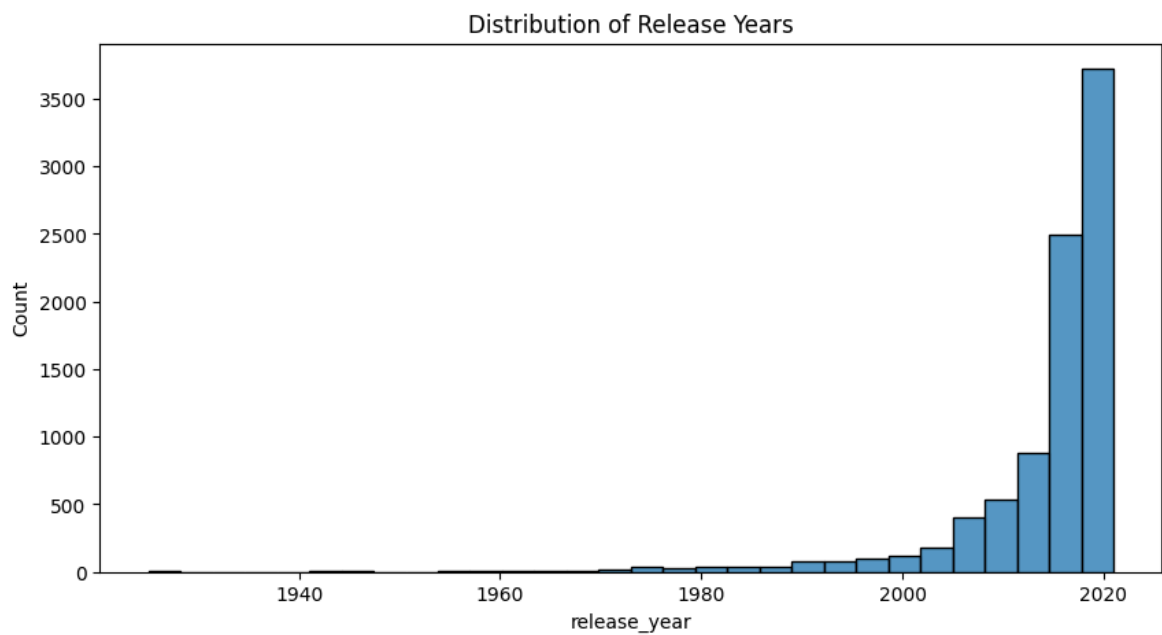
```
In [10]: df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')  
  
df['year_added'] = df['date_added'].dt.year  
  
plt.figure(figsize=(10,5))  
  
df['year_added'].value_counts().sort_index().plot()  
  
plt.title("Content Added Over Years")  
plt.xlabel("Year")  
plt.ylabel("Number of Titles")  
plt.show()
```



```
In [11]: plt.figure(figsize=(10,5))

sns.histplot(df['release_year'], bins=30)

plt.title("Distribution of Release Years")
plt.show()
```

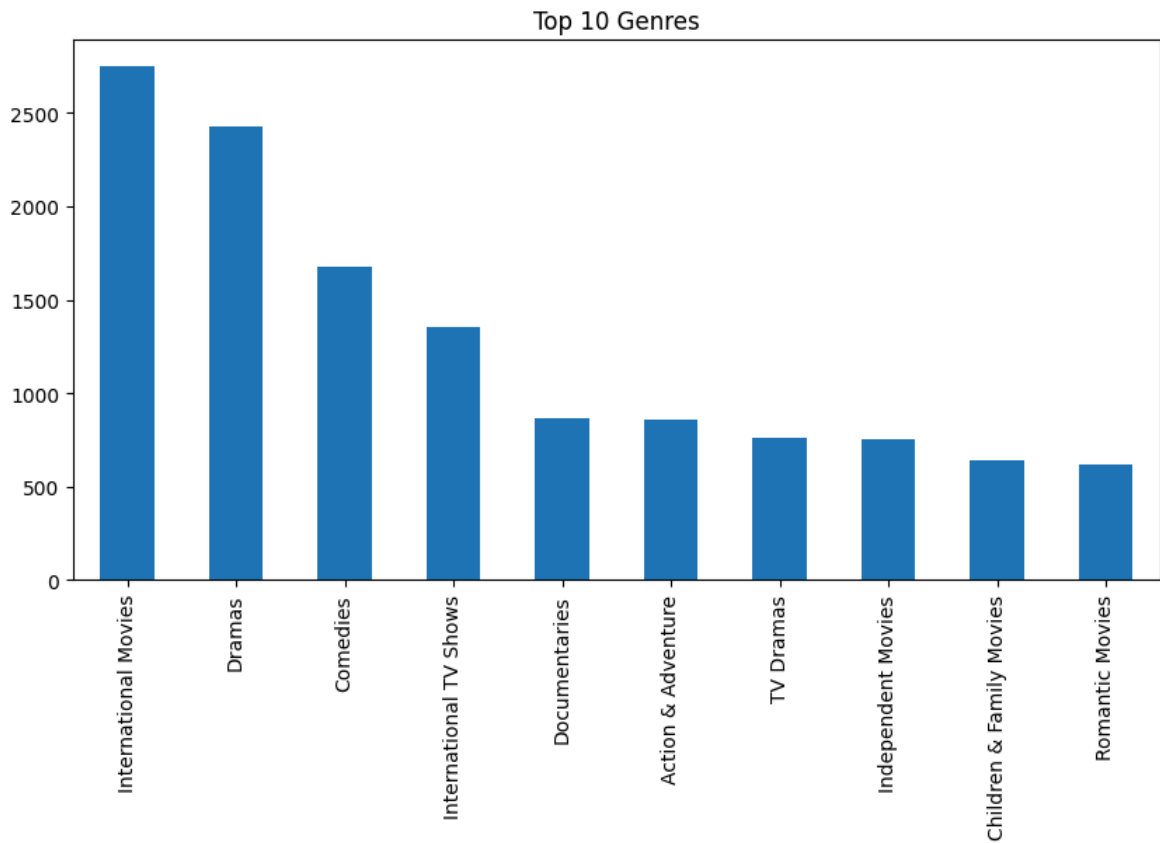


```
In [12]: genres = df['listed_in'].str.split(', ', expand=True).stack()

plt.figure(figsize=(10,5))

genres.value_counts().head(10).plot(kind='bar')

plt.title("Top 10 Genres")
plt.show()
```



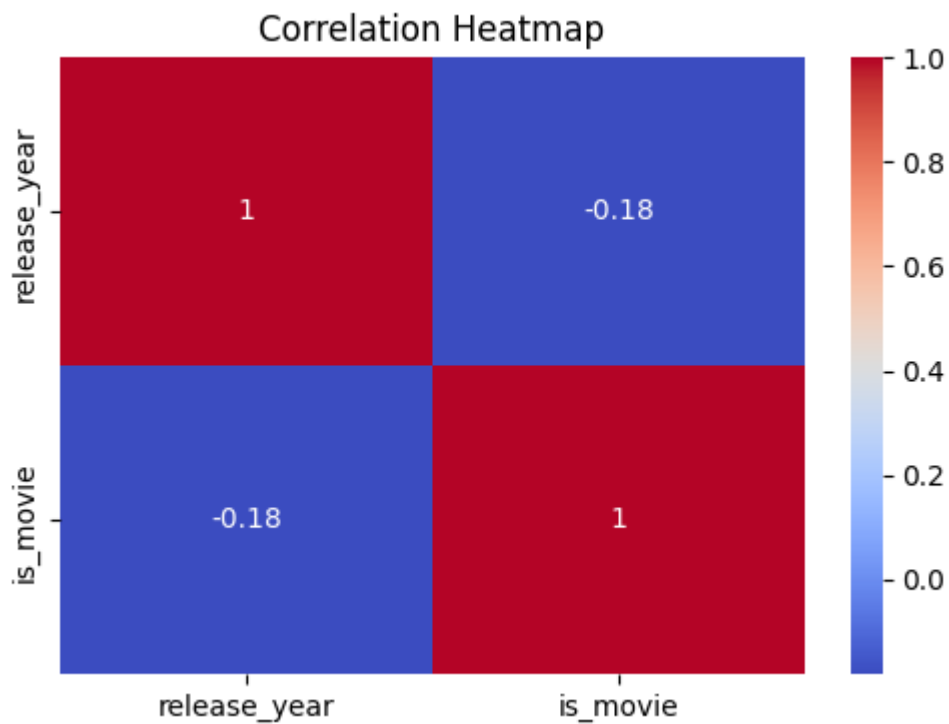
```
In [13]: df['is_movie'] = df['type'].map({
          'Movie':1,
          'TV Show':0
        })

numeric_df = df[['release_year','is_movie']]

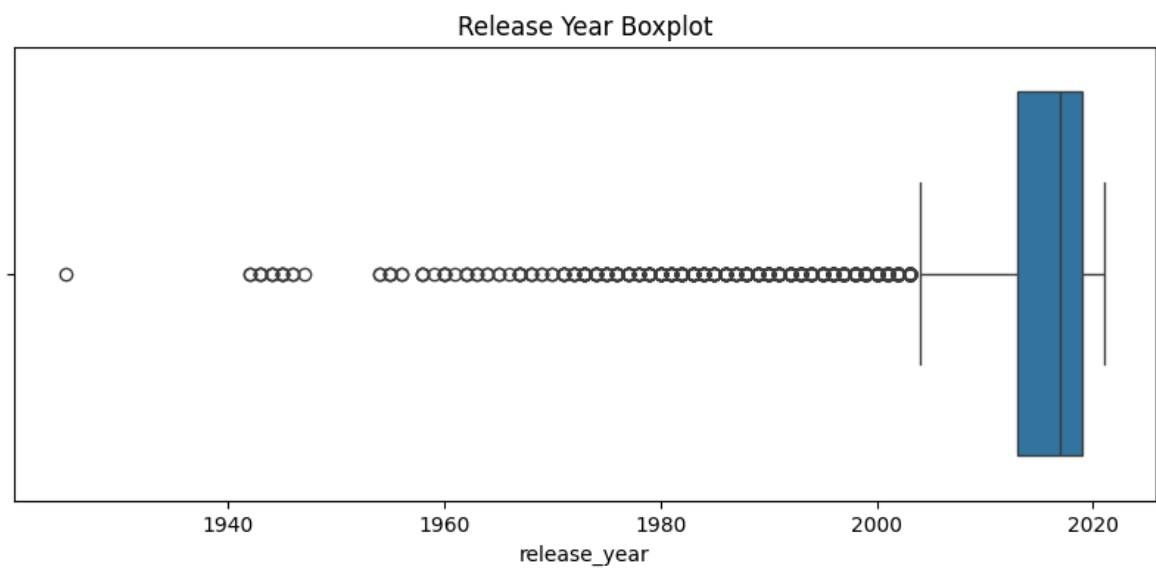
plt.figure(figsize=(6,4))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")
plt.show()
```



```
In [14]: plt.figure(figsize=(10,4))  
sns.boxplot(x=df['release_year'])  
plt.title("Release Year Boxplot")  
plt.show()
```



Summary of Findings

1. Movies significantly outnumber TV Shows.
2. TV-MA and TV-14 are the dominant ratings.
3. The United States produces the highest amount of content.
4. Netflix content additions increased rapidly after 2015.
5. Most titles are from recent years.
6. Drama and International Movies are among the most popular genres.

7. Several fields contain missing values that may require preprocessing.